TRANSFER LEARNING WITH PRETRAINED CNNS FOR IMAGE CLASSIFICATION

Introduction:

The problem of image classification is a seminal task in computer vision in which we try to assign a label to an input image. The traditional way of doing deep learning training convolutional neural networks (CNNs) from scratch requires a large dataset and lots of resources. Nonetheless, in many real-life scenarios, such resources are scarce.

Transfer learning is effective for a task using knowledge learned from a large-scale dataset. Pretrained models leverage previously acquired knowledge to prevent the need for relearning every feature from scratch, leading to quicker training and enhanced performance.

This tutorial investigates the efficiency of transfer learning from pretrained CNNs. The study particularly compares a CNN model trained from scratch as a baseline with a pretrained model used as a feature extractor and a model by fine tuning a pretrained model. The Fashion-MNIST dataset is also used to assess the effects of data augmentation and training efficiency.

Background Theory

Convolutional Neural Networks (CNNs): CNNs perform well on data with a grid-like topology. These include convolutional layers which automatically discover spatial hierarchies of features. The upper layers detect simple patterns and textures, intermediate layers shapes, whereas deeper layers shape object-specific patterns.

The hierarchical learning of these features is quite crucial as it helps us understand the reason behind the success of transfer learning.

Transfer Learning: Transfer learning involves reusing a model trained on a large dataset (e.g., ImageNet) and adapting it to a new task. This reduces the need for large datasets and extensive training.

Two main strategies are used:

- **Feature Extraction:** The pretrained model is frozen, and only new classification layers are trained.
- **Fine-Tuning:** Some of the pretrained layers are unfrozen and retrained to adapt to the new dataset.

Why Transfer Learning Works?

Pretrained models learn general visual features at scale that can be applied to a range of tasks. With the aid of these features, the variance will reduce and thus generalise better.

From a statistical point of view, transfer learning lessens overfitting by restricting the number of parameters to learn from scratch. It is especially helpful when the dataset is small.

Domain Gap and Generalisation: The term domain gap is a significant concept in transfer learning which refers to the difference between source data (ImageNet) and target data (Fashion-MNIST). Although the categories differ, transfer learning still works because low-level features such as edges and textures are universal.

Nonetheless, one may need to fine-tune the higher-level features to better fit the new dataset.

Methodology

Dataset: We used the Fashion-MNIST data set in this study. There are 70,000 grey images of textile in 10 categories. The dataset offers a well-rounded and organized reference for assessing classification models.

Model Configurations:

→Baseline CNN (From Scratch)
A simple convolutional neural network trained without pretrained weights.

→Transfer Learning (Feature Extraction)
A pretrained MobileNetV2 model with frozen layers, used as a feature extractor.

→Fine-Tuned Model
The top layers of the pretrained model were unfrozen and retrained.

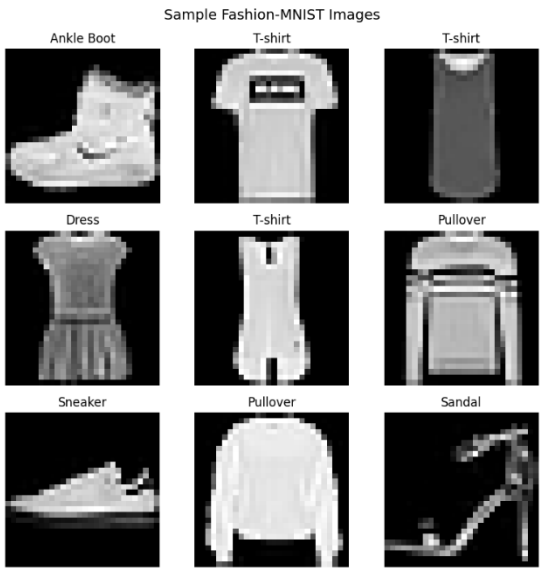
→Transfer Learning with Data Augmentation
Data augmentation techniques such as rotation and flipping were applied to improve generalisation.

Training Setup:

- Optimizer: Adam
- Loss Function: Sparse Categorical Cross entropy
- Epochs: 5
- Metrics: Accuracy and Loss

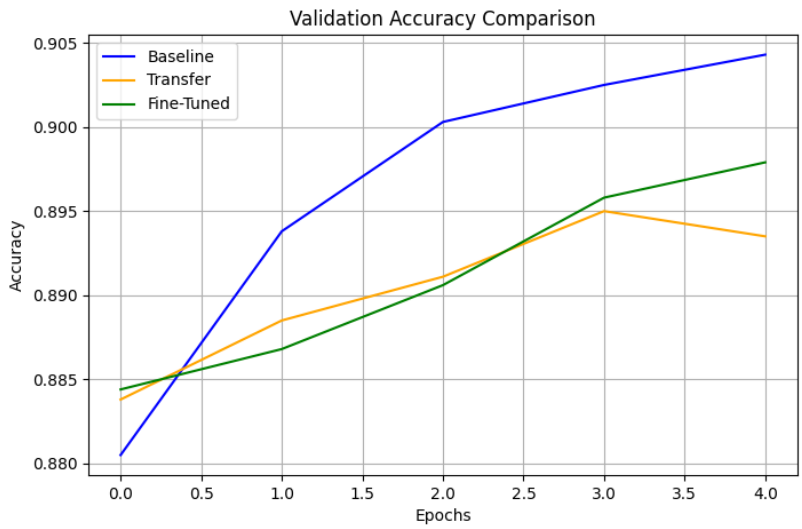
Results and Visual Analysis

Dataset Visualisation: The sample images illustrate the diversity of the Fashion-MNIST dataset. Categories such as shirts and T-shirts appear visually similar, making classification challenging and justifying the need for advanced models.



Accuracy Comparison: The accuracy plot compares the performance of all models across training epochs.

- The base CNN displays a slow and steady improvement.
- The model of transfer learning achieves higher accuracy more quickly, indicating better convergence.
- The fine-tuned model yields the best performance, demonstrating the efficacy of adapting pretrained features.

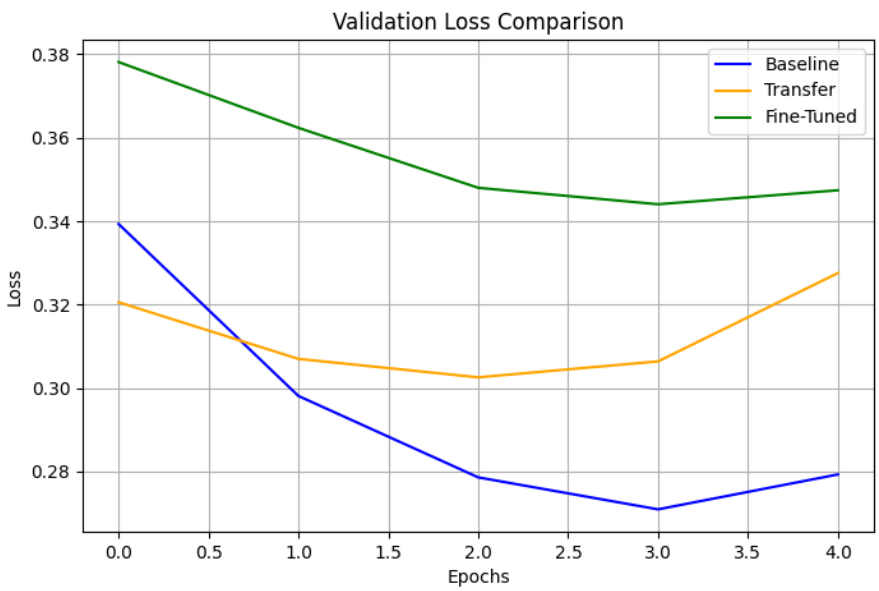


Insight: Transfer learning significantly accelerates learning by leveraging pretrained representations.

Loss Comparison:

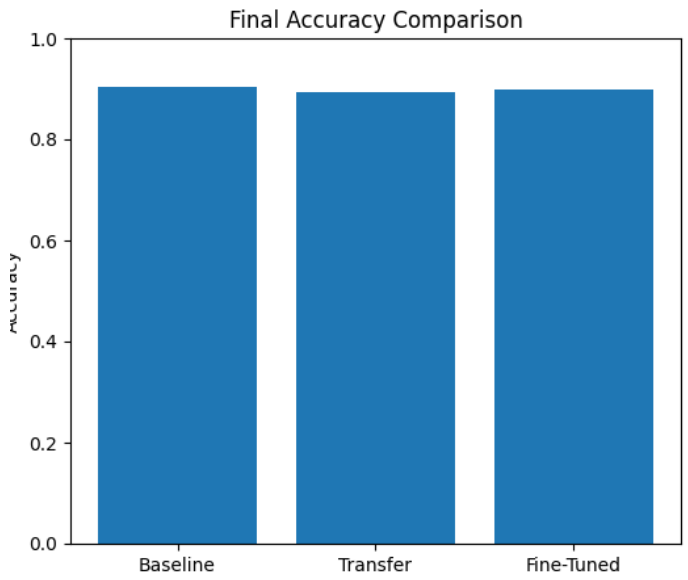
The loss curves reveal differences in training stability.

- The baseline model exhibits higher and more fluctuating loss.
- The transfer learning model shows smoother convergence.
- The fine-tuned model achieves the lowest loss.



Insight: Pretrained models provide more stable optimisation, reducing training instability.

Model Performance Comparison:



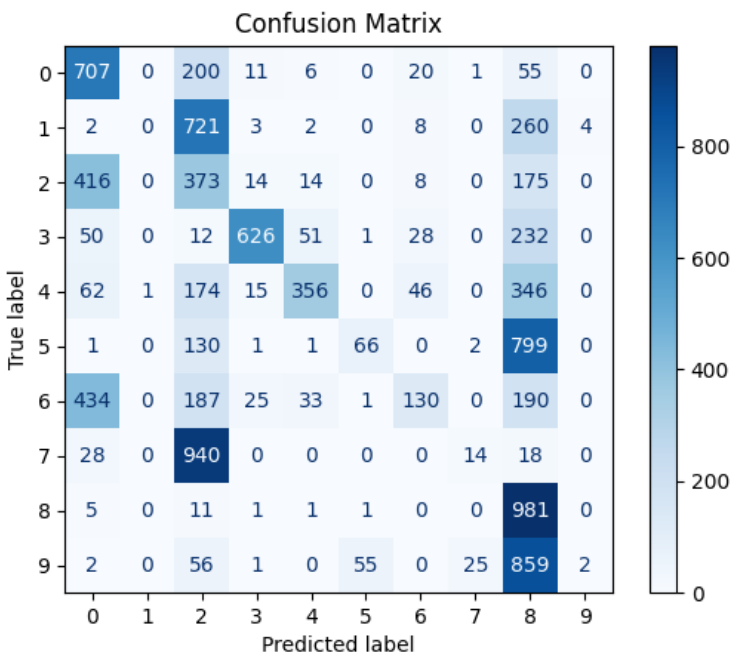
Model	**Performance Insight**
Baseline CNN	Lower accuracy due to limited feature learning
Transfer Learning	Significant improvement due to pretrained features
Fine-Tuned Model	Best performance due to task-specific adaptation

Insight: Fine-tuning enhances performance by refining high-level features.

Confusion Matrix Analysis:

The confusion matrix shows how well each class is predicted.

- Most classes are correctly classified.
- Misclassifications occur between visually similar classes such as shirts and T-shirts.

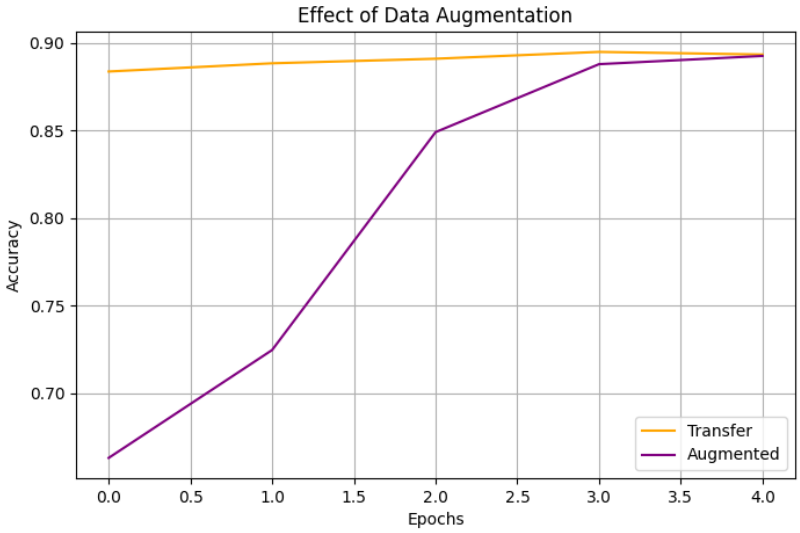


Insight: Even with transfer learning, distinguishing similar classes remains challenging.

Data Augmentation Impact:

The augmentation experiment shows improved accuracy compared to standard transfer learning.

- Augmentation increases dataset diversity.
- It reduces overfitting and improves generalisation.

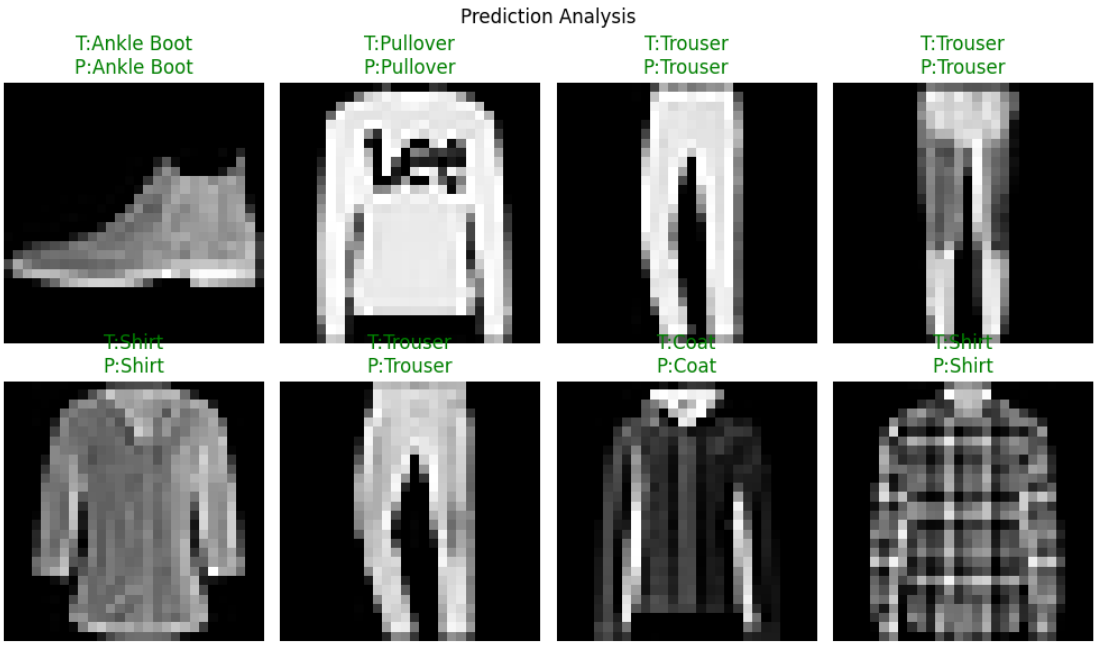


Insight: Data augmentation is a crucial technique for enhancing model robustness.

Prediction Analysis:

Prediction visualisation shows correct and incorrect classifications.

- Correct predictions indicate strong feature learning.
- Errors occur in ambiguous cases.



Insight: Model errors align with dataset complexity, reinforcing earlier findings.

Discussion

The results show that transfer learning gives significant improvement over training CNNs from scratch. Using pretrained models, the network can use more general feature representations, leading to faster convergence and accuracy.

When further fine-tuned, the model can adapt the high-level features to the target data set. The balancing between generalisation and specialisation is necessary for best results. The evidence we have shown that pre-trained models serve well as something new to do.

Data augmentation aids in making model more robust through training. Introducing variation in the training data limits overfitting and allows the model to generalise better to unseen data. It is especially crucial when working with simpler dataset like Fashion-MNIST.

It is also visible from the results that there is a domain gap. Despite the differences between ImageNet and Fashion-MNIST, the implementation of transfer learning remains effective because lower-level features are universal. We need fine-tuning to overcome the gap to higher level representations.

However, other limitations still exist. The model has difficulty with classes that share visual features, indicating the potential need for more sophisticated architectures or an attention mechanism.

Limitations

- Performance depends on similarity between source and target datasets.
- Pretrained models may not generalise well to highly different domains.
- Fine-tuning increases computational cost.
- Fashion-MNIST is simple compared to real-world datasets.
- Misclassification occurs for visually related categories.

Conclusion:

This study shows the benefits of transfer learning for image classification, especially if the resources are limited, including data and computation.

- Transfer learning leads to a significant increase in classification accuracy over a CNN scratch model by employing pretrained models, we gain an advantage in learning speed and effectiveness.
- The findings indicate that by utilizing feature extraction alone, there is a solid baseline produced which leads to higher accuracy and more stable convergence than baseline CNN. This suggests that low-level and mid-level features learned from large datasets are highly transferable.
- When the model is fine-tuned, it only modifies some of the higher-level features which gives more precise results. When implemented, the positive impact on classification accuracy proves the significance of generalisation and specification.
- The use of the data augmenter helps increase generalisation. As a result, overfitting is prevented, and the model performs better on unseen data. This shows that data diversifying is important for practical applications.
- Analysis of the training time shows that transfer learning not only achieves better accuracy but also consumes lesser computational resources and is, therefore, more suited for real-time situations.

- Even after the refinement, the model still makes mistakes in classes that look visually similar for e.g. shirts and T-shirt. Transfer learning is a powerful tool, but it is not sufficient to resolve all ambiguities in a task that is not trivial.
- The domain gap is quite significant because of the difference between the datasets on which the pretrained models are trained. Nevertheless, results do indeed confirm common visual features are transferable across domains.
- The transfer learning for image classification provides a strong and effective approach that can be scaled up in practical deep learning applications.

Future Work

- Apply transfer learning to complex datasets such as CIFAR-10 or real-world images.
- Explore advanced architectures such as EfficientNet.
- Use Grad-CAM for model interpretability.
- Investigate domain adaptation techniques.
- Compare transfer learning with self-supervised learning.

References

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Fashion-MNIST Dataset: https://github.com/zalandoresearch/fashion-mnist